Question:

A baby is born after the 32nd gestational week by cesarean delivery. The mother suffered from gestational diabetes; however, she had no other pregnancy-related diseases and was otherwise healthy. The baby has a blood pressure of 100/58 mm Hg, heart rate of 104/min, and oxygen saturation of 88%. The child has tachypnea, subcostal and intercostal retractions, nasal flaring, and cyanosis. The cyanosis is responding well to initial administration of oxygen. The nasogastric tube was positioned without problems. Which of the following is the most likely diagnosis?

Answer Choices:

A. Neonatal respiratory distress syndrome (NRDS)

B. Tracheoesophageal fistula

C. Pneumonia

D. Congenital heart anomaly with right-to-left shunt

Reasoning + Answer:

<reasoning process here> The baby was born at 32 weeks, which is considered premature, placing the infant at risk for respiratory distress due to underdeveloped lungs and low surfactant production. The clinical signs of tachypnea, subcostal and intercostal retractions, nasal flaring, and cyanosis are classic for respiratory distress. Although oxygenation is improving, these signs are particularly suggestive of neonatal respiratory distress syndrome (NRDS), which is common in preterm infants due to surfactant deficiency. Gestational diabetes may increase the risk of respiratory complications in the neonate due to macrosomia or other factors, but it does not directly cause NRDS.

Tracheoesophageal fistula (B) would typically present with vomiting, feeding difficulties, or aspiration, not primarily with respiratory distress symptoms. Pneumonia (C) would usually have more signs such as fever, crackles on auscultation, and a history of intrauterine infection, and would not typically present with pure cyanosis and retractions without other signs like fever or leukocytosis. A congenital heart anomaly with a right-to-left shunt (D) might cause cyanosis, but in the absence of other signs like a murmur or decreased cardiac output, and especially with a normal blood pressure and heart rate, this is less likely.

The established clinical presentation matches the typical features of neonatal respiratory distress syndrome, especially in a preterm neonate. The lack of other systemic issues and the presence of tachypnea and retractions align with NRDS.

<answer> A. Neonatal respiratory distress syndrome (NRDS) </answer>